

Boao (James) Chen

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Research Interests

Databases, Data Systems, Storage Engines, Machine Learning for Systems

Education

Doctor of Philosophy, Computer Science

Brandeis University, Waltham, MA

Advisor: Subhadeep Sarkar

Sep 2024 – Present

Bachelor of Arts, Computer Science

Cornell University, Ithaca, NY

Washington University, St. Louis, MO

Aug 2020 – May 2024

Publications

- **Boao Chen**, Shubham Kaushik, and Subhadeep Sarkar, “[Mind the Buffer: Dissecting the LSM-Buffer Design Space](#)”, *in Review*, 2026
- Alexander H. Ott, Shubham Kaushik, **Boao Chen**, and Subhadeep Sarkar, “[Tectonic: Bridging Synthetic and Real-World Workloads for Key-Value Benchmarking](#)”, *Performance Evaluation and Benchmarking TPC Technology Conference*, 2025
- Ke Li, Ruidong Zhang, **Boao Chen**, Siyuan Chen, Sicheng Yin, Saif Mahmud, Francois Guimbretiere, and Cheng Zhang, “[GazeTrak: Exploring Acoustic-based Eye Tracking on a Glass Frame](#)”, *International Conference on Mobile Computing and Networking*, 2024
- Ke Li, Ruidong Zhang, Siyuan Chen, **Boao Chen**, Mose Sakashita, Francois Guimbretiere, and Cheng Zhang, “[EyeEcho: Continuous Facial Expression Tracking on Glasses with Minimally-obtrusive Acoustic Sensing](#)”, *ACM Computer Human Interaction conference*, 2024
- Ke Li, Devansh Agarwal, Ruidong Zhang, Vipin Gunda, Tianjun Mo, Saif Mahmud, **Boao Chen**, Francois Guimbretiere, and Cheng Zhang, “[SonicID: User Identification on Smart Glasses with Acoustic Sensing](#)”, *The Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 2024

Poster

- Shubham Kaushik, Steven Yang, **Boao Chen**, and Subhadeep Sarkar, “[FluidLSM: A Fluid Log-Structured Merge Tree for Shifting Workloads](#)”, 2026 North East Database Day.
- Alexander Ott, Shubham Kaushik, **Boao Chen**, and Subhadeep Sarkar, “[Tectonic: Bridging Synthetic and Real-World Workloads for Key-Value Benchmarking](#)”, 2026 North East Database Day.
- Alexander Ott, **Boao Chen**, Arpita Saha, and Subhadeep Sarkar, “[Self-Designing LSM Memory Buffer](#)”, 2025 North East Database Day.

Teaching Assistantship

- **Spring 2026**: *Introduction to Problem Solving in Python* [COSI 10a](#), Brandeis University.
- **Fall 2025**: *Capstone Project for Software Engineering* [COSI 166B](#), Brandeis University.
- **Spring 2025**: *Database Management Systems* [COSI 127B](#), Brandeis University.
- **Fall 2024**: *Algorithms* [COSI 180A](#), Brandeis University.
- **Spring 2024**: *Applied High-Performance and Parallel Computing* [CS 5520](#), Cornell University.
- **Fall 2023**: *Foundations of Artificial Intelligence* [CS 4700](#), Cornell University.
- **Spring 2023**: *Operating Systems* [CS 4410](#), Cornell University.

Certification

- **June 2025:** *Oracle Certified Professional, MySQL 8.0 Database Administrator* — [\[link\]](#)
- **June 2025:** *Oracle Certified Associate, MySQL Implementation* — [\[link\]](#)

Technical Skills

- **Programming Languages:** Python, C/C++, Java, SQL, R, MATLAB
- **Databases:** RocksDB, MariaDB, Postgres, Cassandra
- **Tools:** PyTorch, TensorFlow, OpenCV

Society Member

Member, Association for Computing Machinery (ACM)